

Henry Demarest

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Education

Yale University: New Haven, Connecticut (2021 to Present)

- *Intended Majors:* BS Mechanical Engineering (ABET), BS Computer Science | *GPA:* 3.96
- *Relevant Coursework:* Thermodynamics, Linear Algebra, Differential Equations, Fluid Mechanics, Material Science, Statics, Computer-Aided Design, Mechatronics, Mechanical Design, Data Structures, Computer System Architecture, Algorithms
- *Non-Technical Activities:* Yale Club Running Team Captain, Violist in DPops Orchestra, Engineering Tour Guide

Irvington High School: Irvington, New York (September 2017 to June 2021)

- Weighted GPA: 4.58, SAT: 1600
 - *Awards:* Valedictorian (2021), National Merit Semifinalist (2021), U.S. Presidential Scholar Semi-Finalist (2021), USA Computing Olympiad Gold Status (2019), American Computer Science League National Gold Medal (2021)
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Experience

Yale Undergraduate Aerospace Association (2021 - Present):

Project Liquid – Co-President/Technical Lead (2022-Present), Thrust Chamber Subteam Lead (2021 - 2022):

- Lead technical development of a 200 lbf bipropellant liquid rocket engine designed for use with Nitrous Oxide and IPA
- Coordinate and assist subteams to ensure design/manufacturing progress meets deadlines and technical objectives
- Teach others about concepts related to fluid mechanics, 3D design, and rocket engine operation
- Collaborate with similar teams at other schools, research previous engine designs, and establish documentation standards
- Led design of a coaxial shear injector that uses a stacked manifold design as Thrust Chamber Subteam Lead

CubeSat Team – Attitude Determination and Control System(ADCS) Lead (2022 - Present), Member (2021-2022):

- Lead the ADCS Subteam to prepare CubeSat for launch as part of NASA's CubeSat Launch Initiative Award
- Design algorithm to control CubeSat's active stabilization efforts (see research below)
- Collaborated on the design of an attachment system for the Gravity Gradient Boom (GGB)

Yale University – Undergraduate Learning Assistant for Computer Science Course (2022 - Present):

- Work as a member of the teaching staff for CPSC 223 (Data Structures & Programming Techniques)
- Hold office hours and teach students topics ranging from specific programming syntax to general problem solving strategies

First Robotics Competition Team – Captain (2020-2021), Lead Programmer (2018 - 2020):

- Designed, CAD modeled, and constructed robots to compete in the First Robotics Competition
- Programmed autonomous and teleoperated modes for improved performance and intuitive control
- Wrote grants, met with local organizations, and coordinated with school administration

Research

Cube Satellite Detumbling Research Project – Researcher (Summer 2022):

- Spent 10 weeks working on a detumbling algorithm for a 2U CubeSat with a limited suite of sensors and control hardware
- Used TRIAD algorithm for attitude determination and B-dot controllers paired with magnetorquers for attitude correction
- Created MATLAB simulations to experiment with various control system parameters and assess algorithm performance

Machine Learning Science Research Project – Student Researcher with Mentors at IBM (2019 - 2021):

- Wrote Python code to assess how well different reinforcement learning algorithms can train a bipedal model to walk
- Earned 1st Place in Computer Science Category (2021) at the NYS Science and Engineering Fair (NYSSEF)
- Competed at the Regeneron International Science and Engineering Fair (ISEF) in 2021

Volunteering

Code Haven – Student Mentor (2021 - 2022):

- Taught coding skills to New Haven middle school students, explaining complex topics in fun and understandable ways

Fuller Center for Housing – High School Chapter Founder, Volunteer Supervisor, Intern (2018 - 2021):

- Repaired and remodeled local homes to be used as affordable housing in local communities
 - Volunteered for up to 35 hours/week with this non-profit during the summers and founded a chapter at my high school
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Skills

Computer Programming: C, Java, Python, MATLAB

CAD: Solidworks, Autodesk Inventor, Autodesk Fusion360

Manufacturing: Lathe, Mill, CNC, 3D-Printer, Hand Tools

Software: Project Organization (ClickUp, Slack), Project Sharing (GrabCAD, GitHub), Microsoft Office

Language: Conversational Spanish